

Justin Mai

justimai042@gmail.com | (626) 271-7071 | Los Angeles, CA | [Website](#) | [GitHub](#)

EXPERIENCE

GeoCam - Data Science Intern (Capstone)

Jan 2026 - Jun 2026

GeoCam Project Team

San Luis Obispo, CA

- Built Python and DuckDB workflows to query geospatial data directly from AWS S3 using SQL, utilizing ETL pipelines and database querying to transform raw GeoCam captures into analysis-ready datasets containing 26,000 camera-pose records across 8,600 image captures.
- Fine-tuned an OpenCLIP model on GeoCam image data and integrated LightGlue feature matching, developing a machine learning pipeline to retrieve visually similar reference images and predict campus locations.
- Generated reproducible geospatial datasets and visualization outputs for 3D Gaussian Splatting scene reconstruction and location detection for in development Cal Poly virtual touring application.

Research Intern

May 2025 - Sep 2025

Cal Poly, College of Science & Mathematics

San Luis Obispo, CA

- Built a reproducible R/tidymodels statistical modeling and machine learning pipeline benchmarking 12 regression algorithms across 32 synthetic datasets.
- Conducted statistical analysis of model performance using R-squared, Friedman tests, and post-hoc Wilcoxon signed-rank tests to identify significant performance differences across regression algorithms.
- Produced a manuscript, poster, and technical talk translating model performance results for statistical and nontechnical audiences.

Independent Analyst

Jul 2024 - Present

Sports Analytics and Market Modeling

- Built R/Python models using quarterback, receiver, matchup, and market data to identify pricing inefficiencies in NFL markets.
- Performed data cleaning, feature selection, model evaluation, and statistical modeling for forecasting.
- Tracked model performance across historical outcomes.

TECHNICAL SKILLS

- **Languages:** SQL, Python, R, Java, SAS
- **Analytics and modeling:** regression, Bayesian methods, time series, mixed-effects models, multivariate analysis, experimental design, sampling, feature engineering, cross-validation, model evaluation
- **Libraries and tools:** PowerBI, pandas, scikit-learn, tidyverse, tidymodels, R, Excel, Shiny, Jupyter/Colab, Git, GitHub, AWS
- **Data workflow:** database querying, data cleaning, reproducible pipelines, dashboards, technical documentation, statistical communication

EDUCATION

Cal Poly San Luis Obispo

Graduated: June 2026

Bachelor of Science in Statistics; Minor in Data Science

San Luis Obispo, CA

- **Relevant coursework:** Advanced Statistical Computing in R, Bayesian Reasoning, Knowledge Discovery from Data, Database Systems, Data Science Capstone, Time Series, Multilevel and Mixed Modeling, Multivariate Statistics, Applied Linear Models, Experimental Design, Linear Algebra.

PROJECTS

- **Census Visualizer:** Deployed an interactive R Shiny web application integrating the U.S. Census API to visualize population and income trends across 3,000+ U.S. counties, with dynamic filtering by race, year range, and demographic measure.
- **DataFest 2025:** Built a student-similarity analysis for CourseKata textbook data in R/Python, using performance matrices and cosine similarity to compare learning patterns.